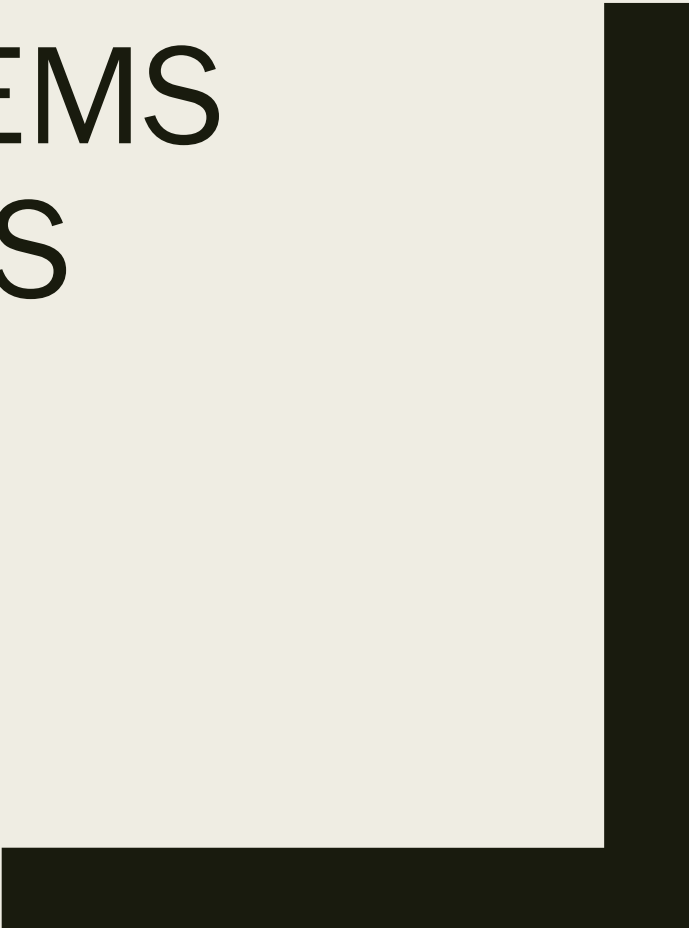




COMPUTER SYSTEMS FUNDAMENTALS (4COSC004W)

Lecture: 11 Nov



In this lecture we will cover:

- Network topologies
 - *Physical & Logical*
- Types of network
- Network components
- IP Addressing
 - *Calculations*
 - *Masking*
 - *Classless & Classful systems*
- Subnetting calculations

NETWORK TOPOLOGIES

Physical & Logical Topologies

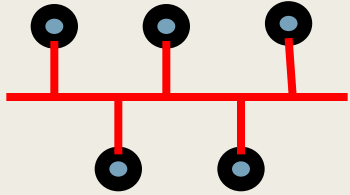
Networked computers

- Computers connected to each other
 - *Wired connections using cables*
 - *Wireless using WIFI, 4G/5G*
- We will concentrate on wired arrangement

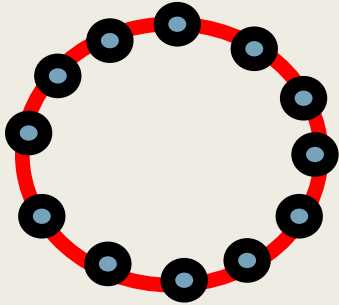
Physical & Logical Topologies

- **Physical Topologies** define the actual layout of the wire (media)
 - *eg. The wiring in the computer labs are laid out in an extended star arrangement*
 - ***Terminology:*** *Network Interface Card (NIC), Unshielded Twisted Pair (UTP), Shielded Twisted Pair (STP)*
- **Logical Topology** defines how the media is accessed by the hosts
 - *eg. In the computer labs, hosts access the media on a first come, first served basis*

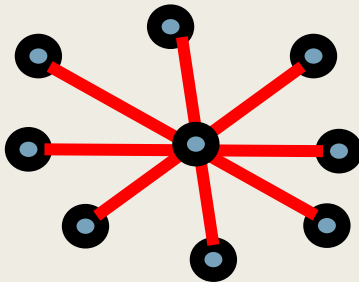
Physical Topologies



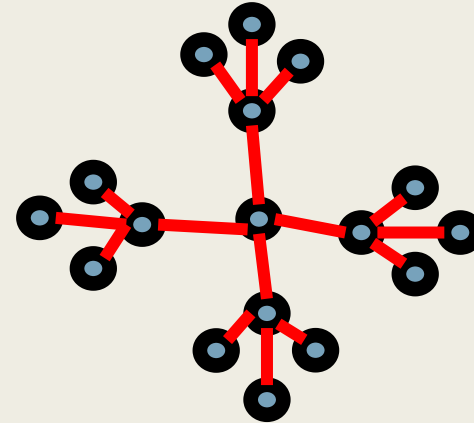
Bus



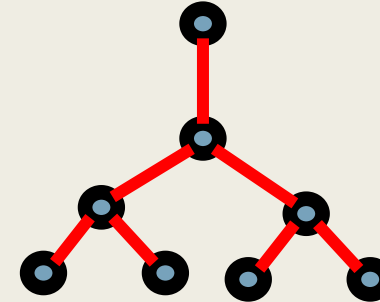
Ring



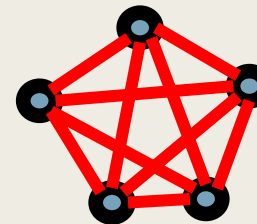
Star



Extended Star



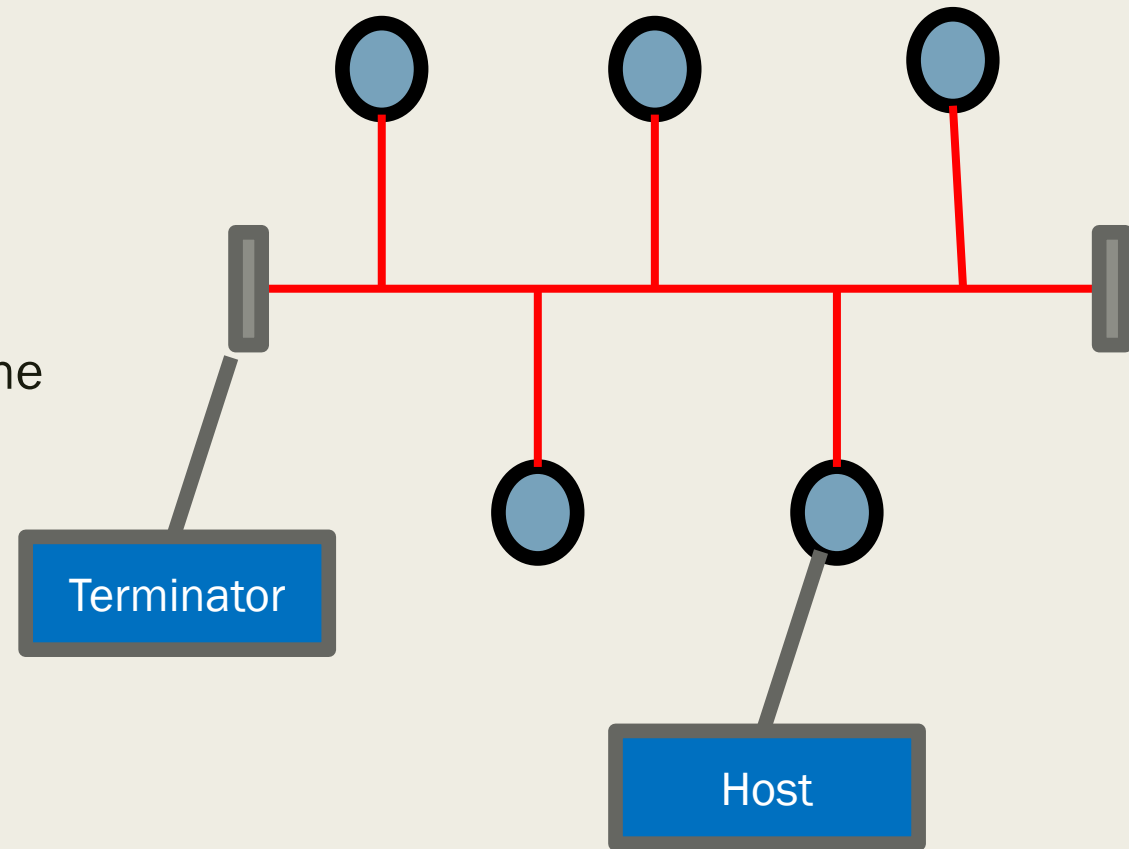
Hierarchical



Mesh

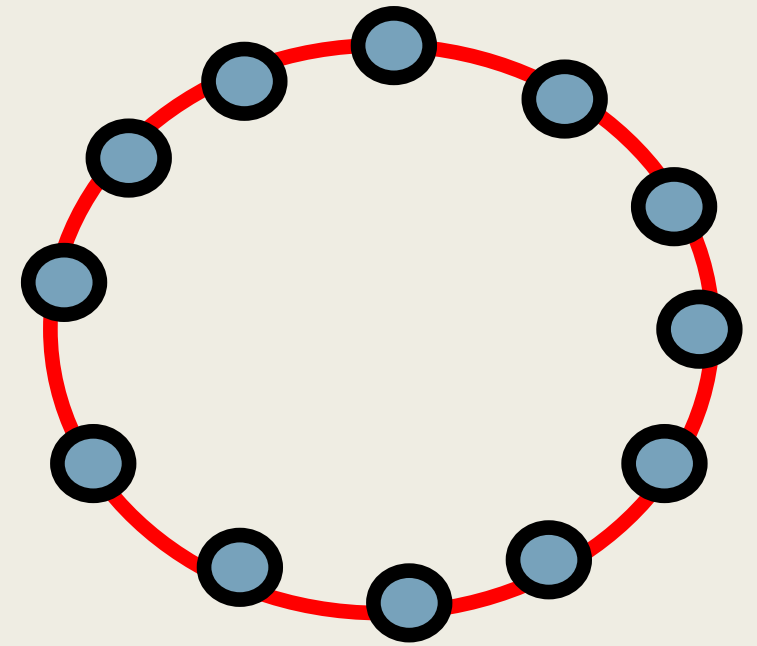
Bus topology

- Single backbone
- All hosts connected to the backbone
- Each end must be terminated
- Susceptible to collisions



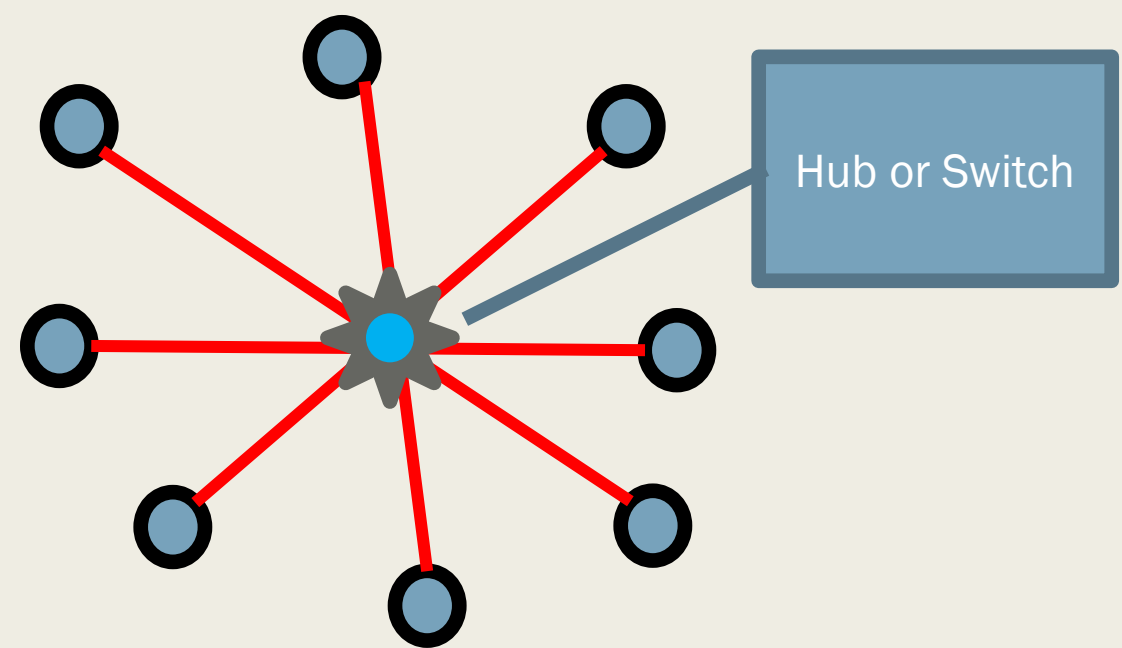
Ring Topology

- No Backbone
- A host is directly connected to each of its neighbours
- Used for Token Passing logical topologies



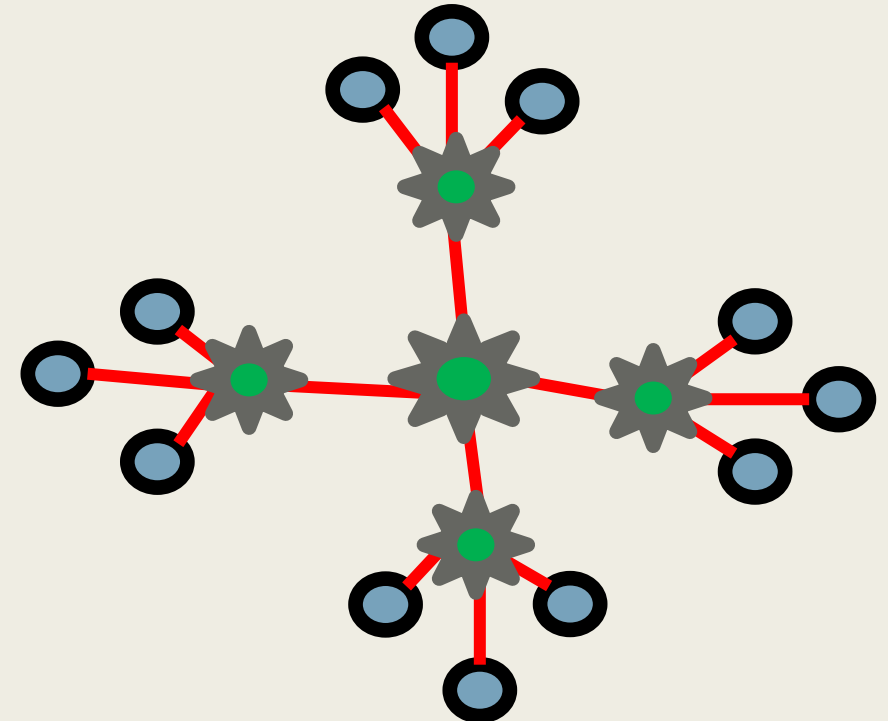
Star Topology

- All devices connected to a central point
- Used for Ethernet technologies



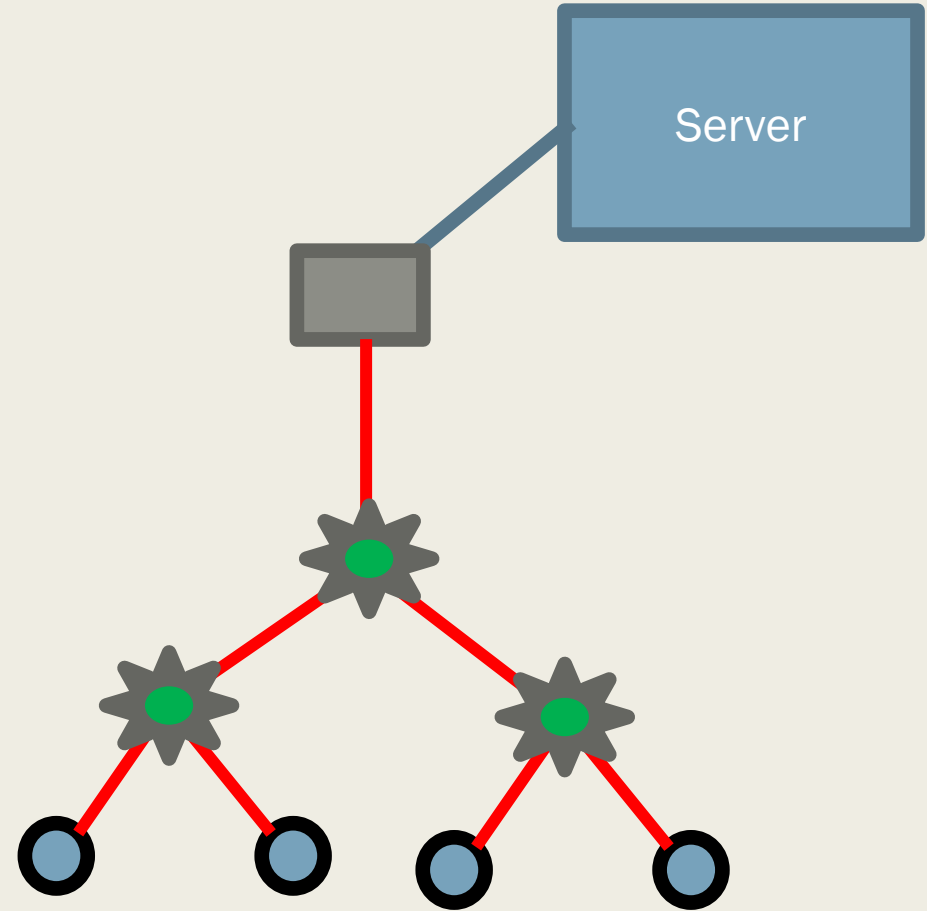
Extended Star Topology

- Connects Star topologies together
- Fractal pattern
- At the centre of Star is a Hub or Switch
- Extends the size of the network



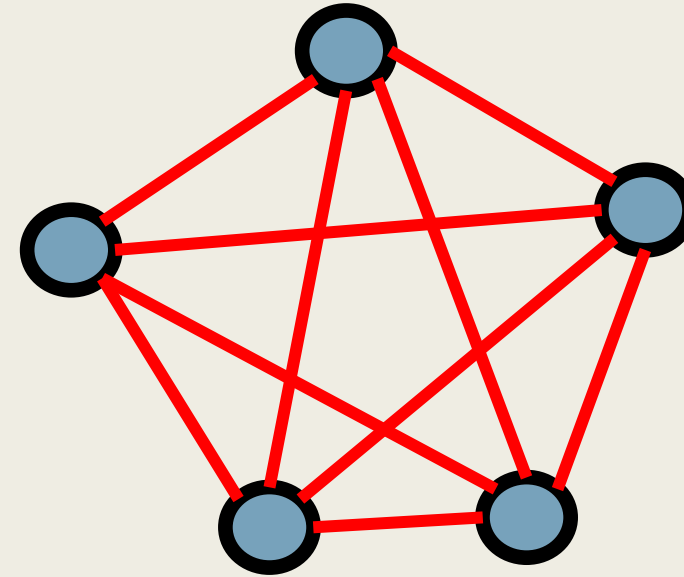
Hierarchical topology

- Like the extended star
- Except a computer controls traffic
 - *NOT a Hub or Switch*



Mesh Topology

- Maximally connected:
 - *Each host has its own connection to every other host*
 - *Use for critical systems*
- Non-maximally connected:
 - *Not every host is connected to every other host*
 - *Alternate routes if there are problems*



Logical Topologies:

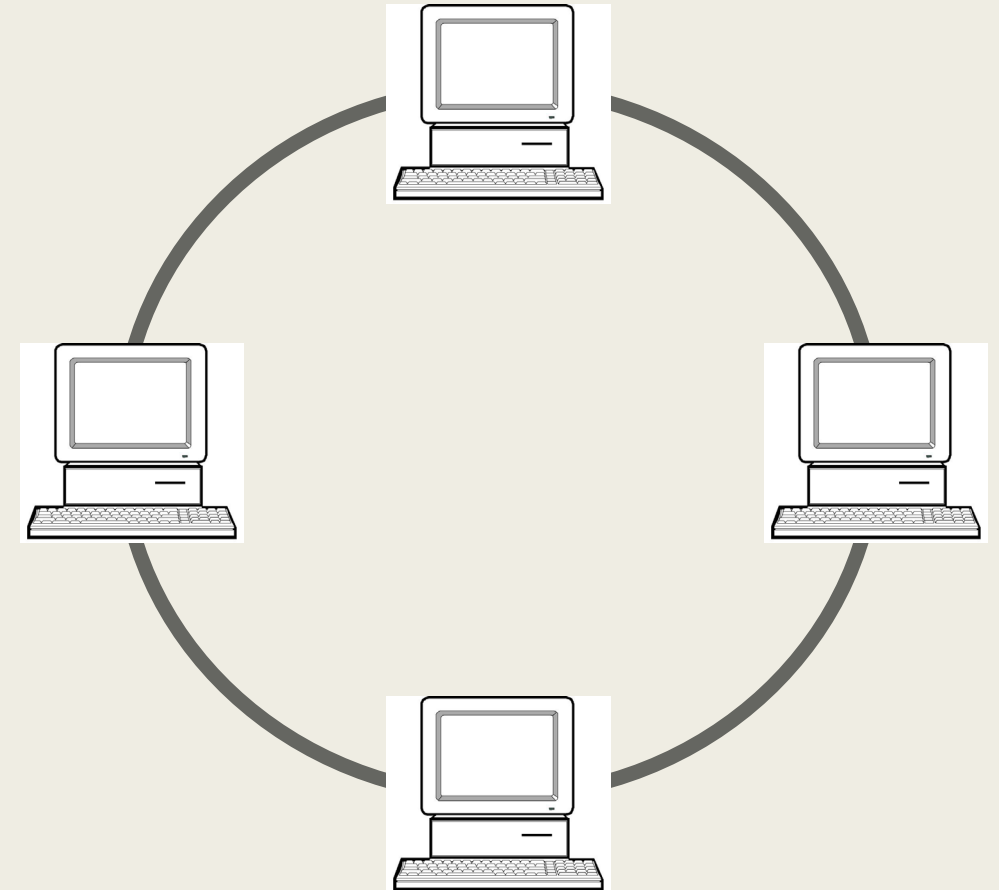
- Broadcast topology:
 - *Each host on the LAN sends (or broadcasts) its data to every other host.*
 - *Access to media is based on “First come, first served”*
 - *Ethernet works this way*
- Token Passing Topology:
 - *Access to media is controlled by an electronic token*
 - *Possession of the token gives the host the right to pass data onto the media.*

TYPES OF NETWORK

Peer-to-peer
Client-Server
Sizes of networks

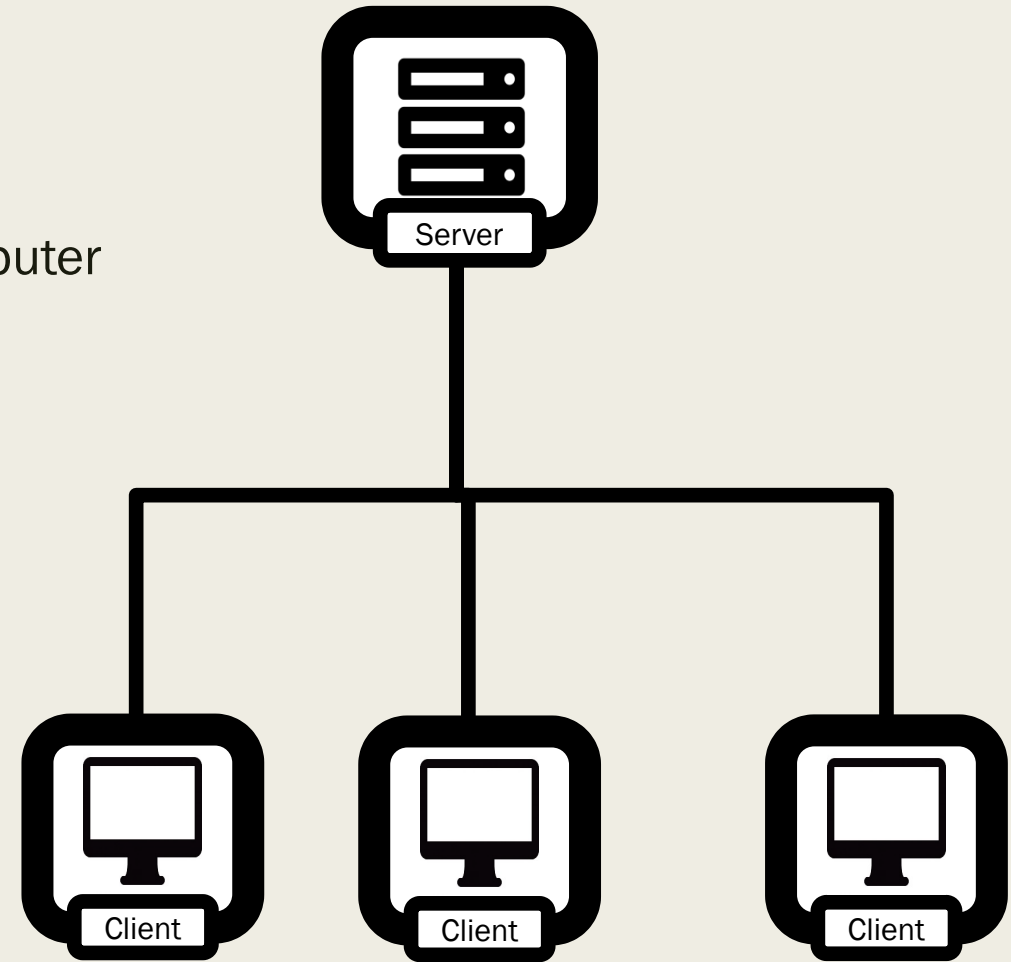
Peer-to-Peer Network

- Networked computers are equal partners
- Each computer can be a Server or Client
- Each component controls its own resources
- Resources can be shared
- Suitable for small networks



Client/Server network

- Network services are located on a dedicated computer
 - *The Server*
- Server responds to requests from **Clients**
- Resources are shared
- Server can serve many Clients simultaneously
- Needs an administrator



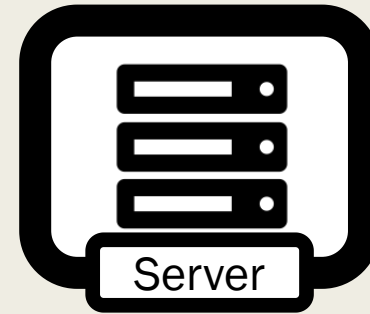
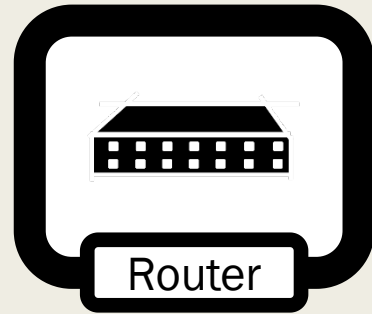
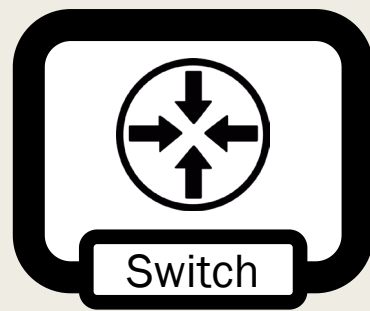
Terminology: Sizes of networks

- LAN
 - *Local Area Network*
- WAN
 - *Wide Area Network*
- MAN
 - *Metropolitan Area Network*
- SAN
 - *Storage Area Network*

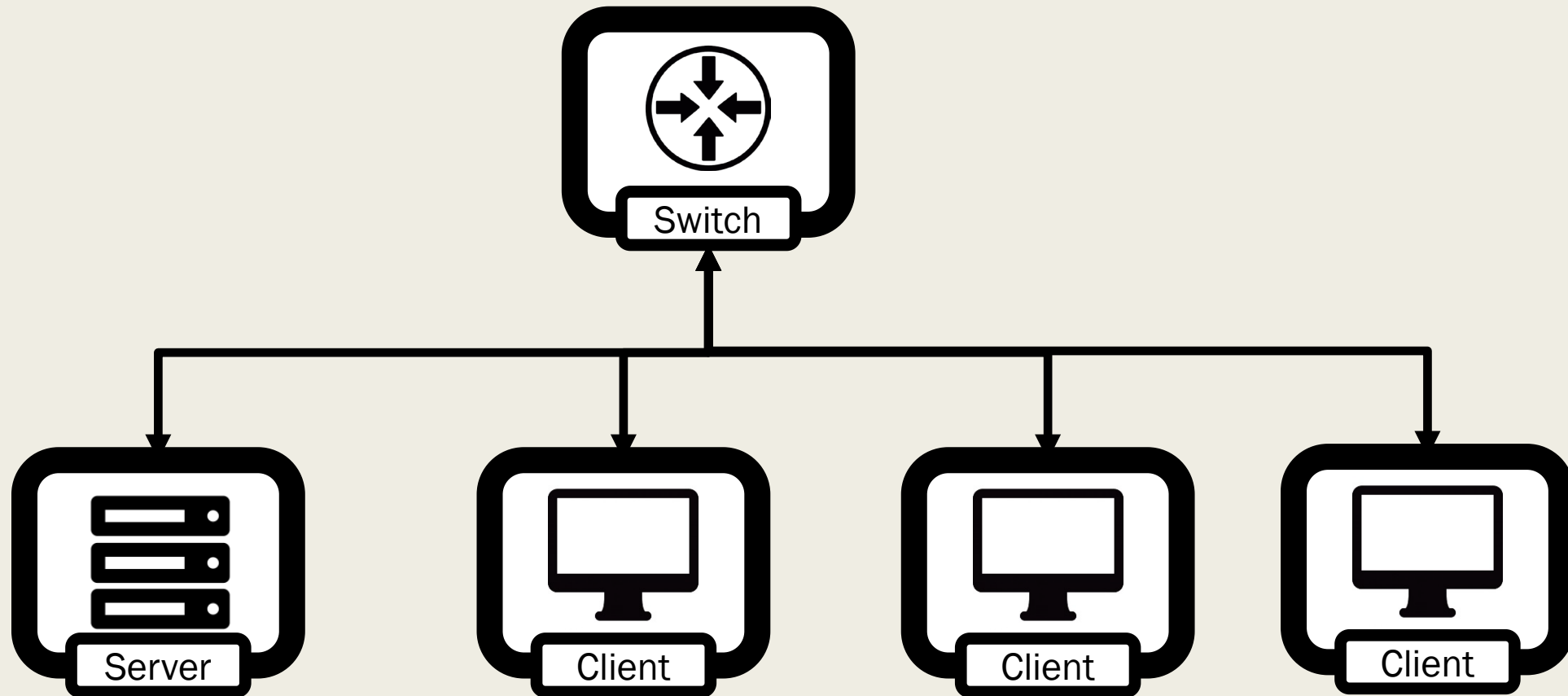
NETWORK COMPONENTS

Actors on the Network

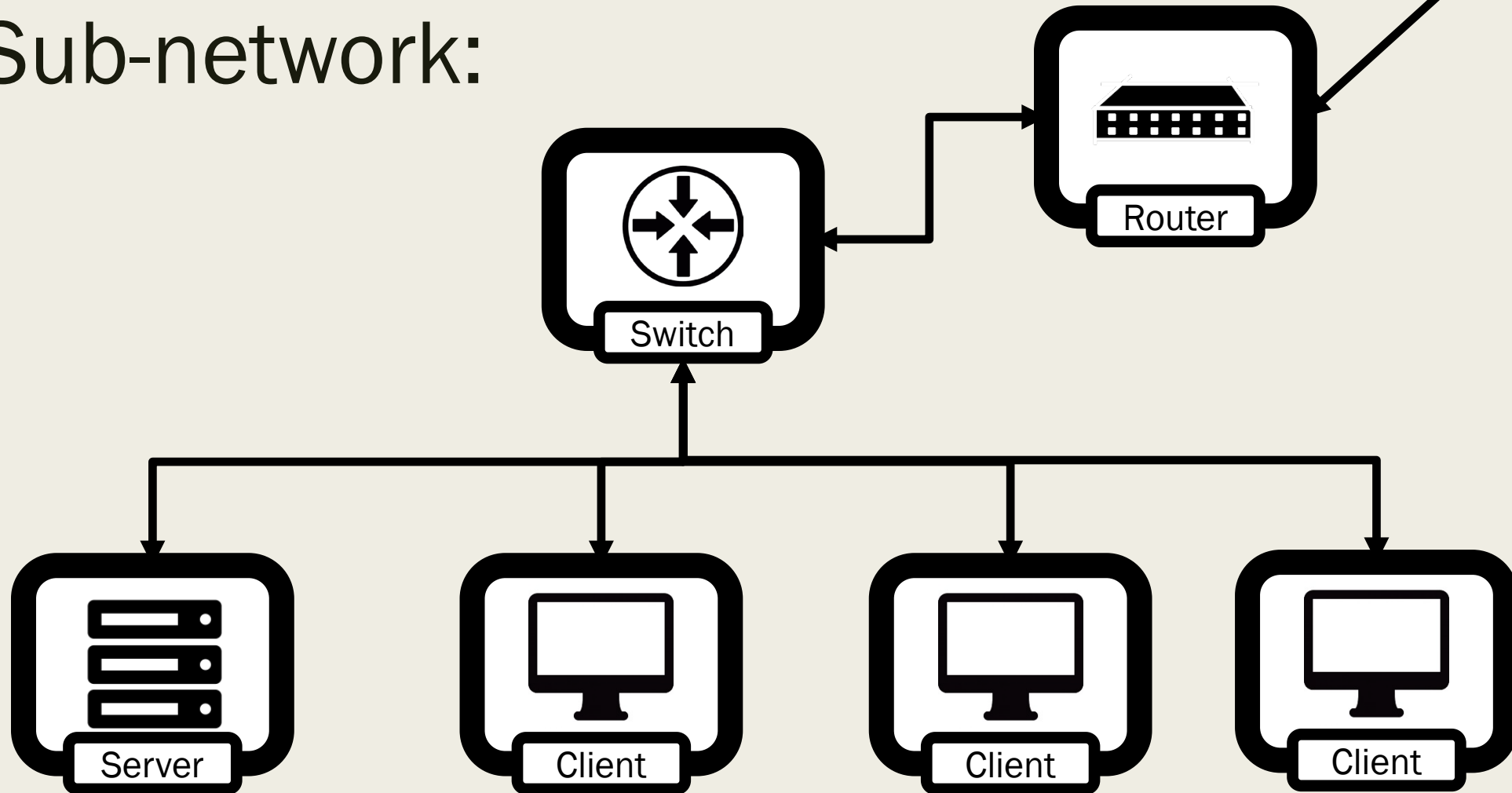
Network components:



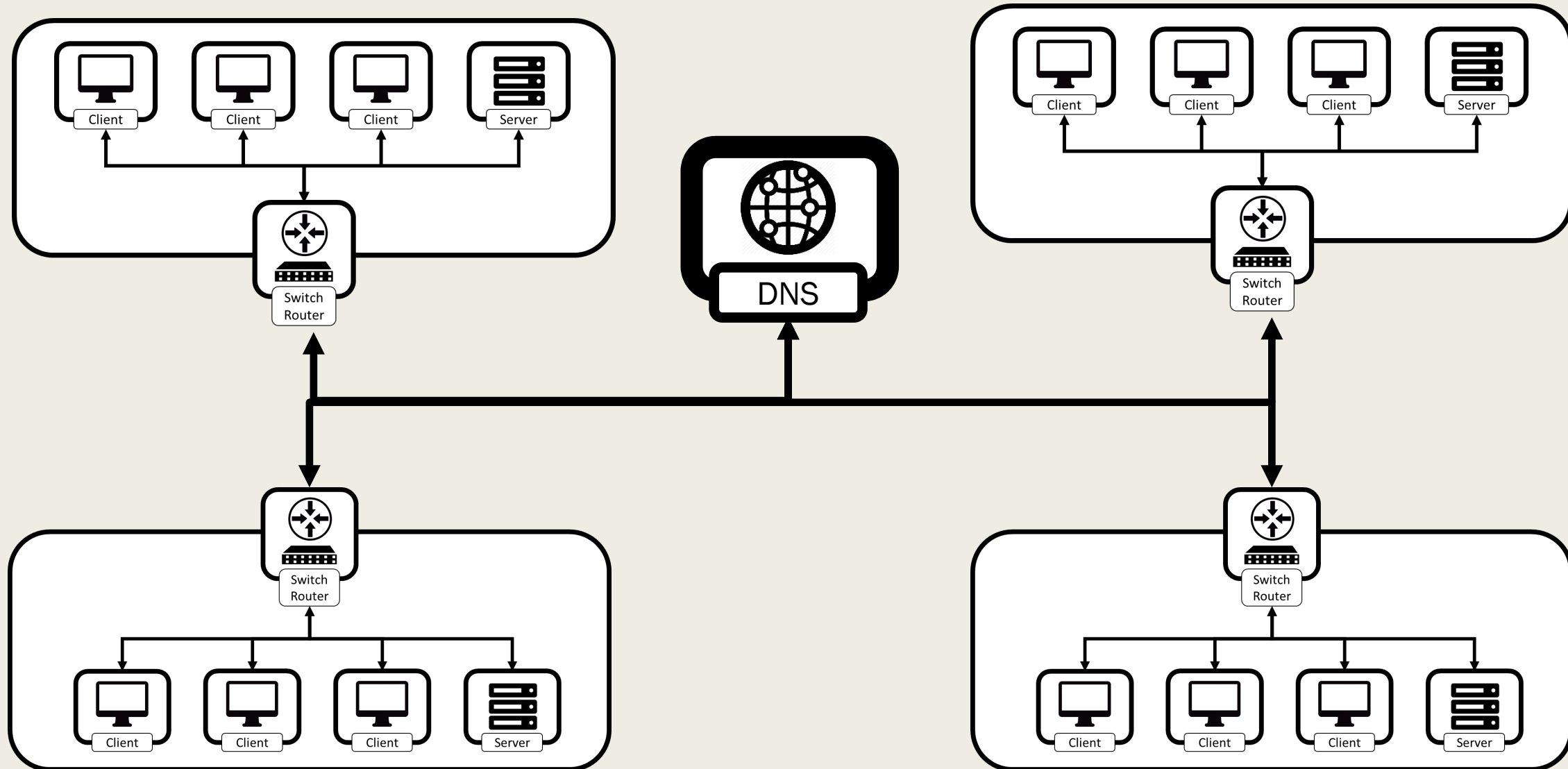
Closed network:



Sub-network:



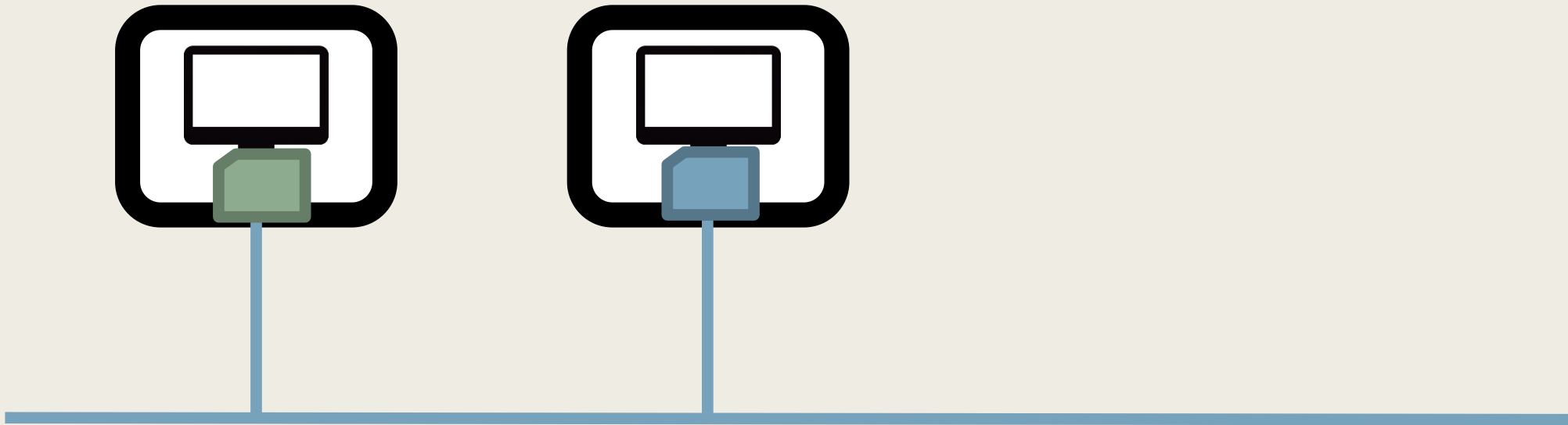
Complete network:



NETWORK COLLISIONS

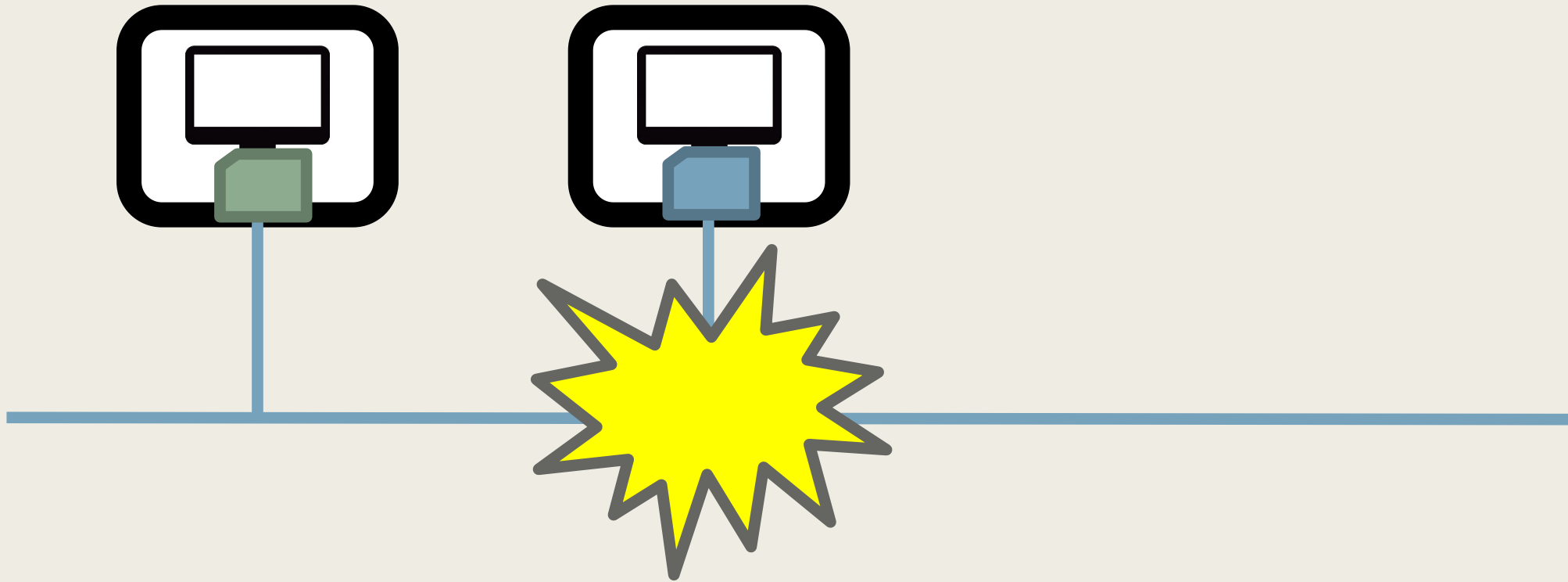
Contention-Based Network Access

Collisions : non-collision scenario



- Occur in Broadcast topologies where nodes use shared media

Collisions : collision scenario



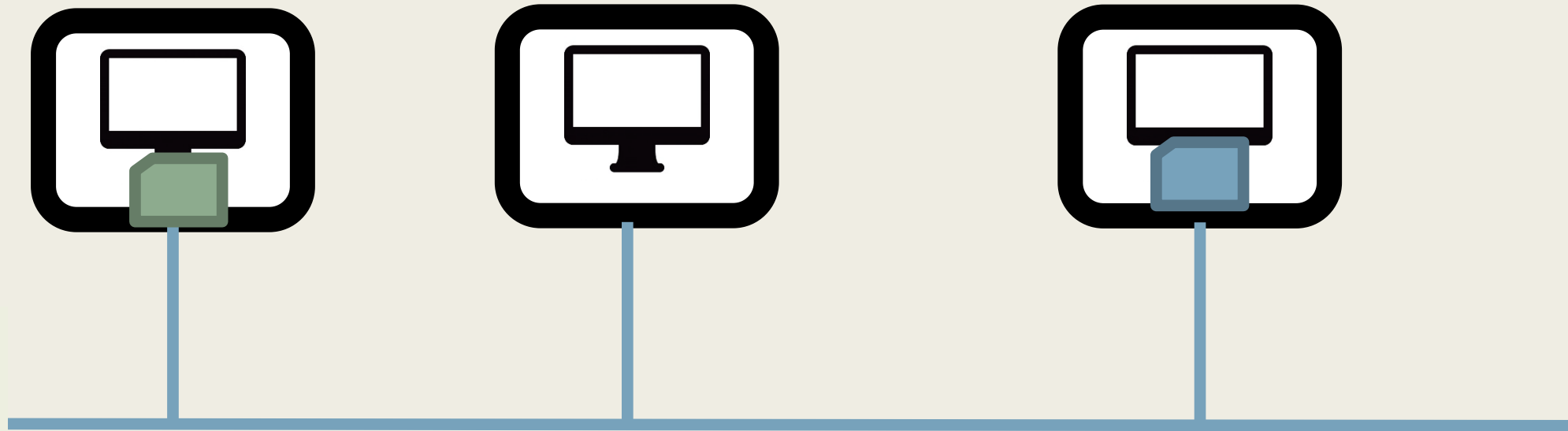
- Occur in Broadcast topologies where nodes use shared media

Contention-Based Network Access

- Carrier Sense Multiple Access / Collision Detection
 - CSMA/CD
 - *Commonly used in Wired Ethernet*
- Carrier Sense Multiple Access / Collision Avoidance
 - CSMA/CA
 - *WiFi*

CSMA/CD

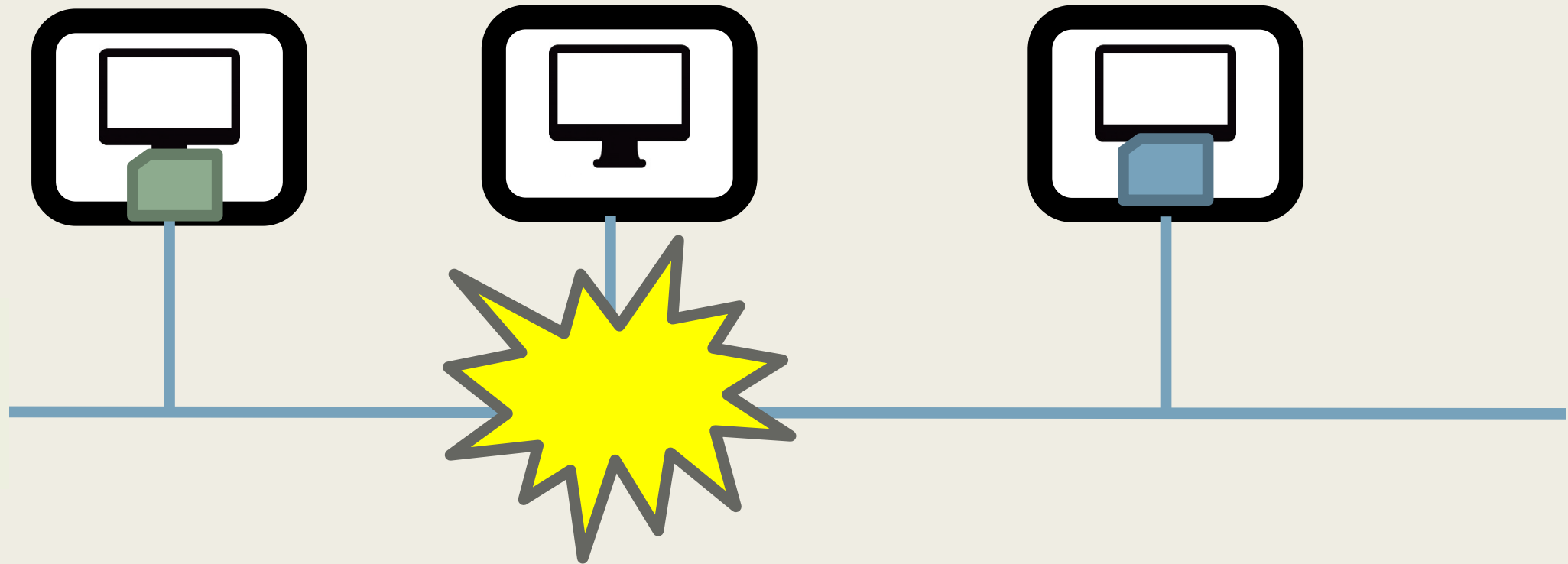
Carrier Sense Multiple Access / Collision Detection



- Sending node will wait till it can hear no traffic on the network

CSMA/CD

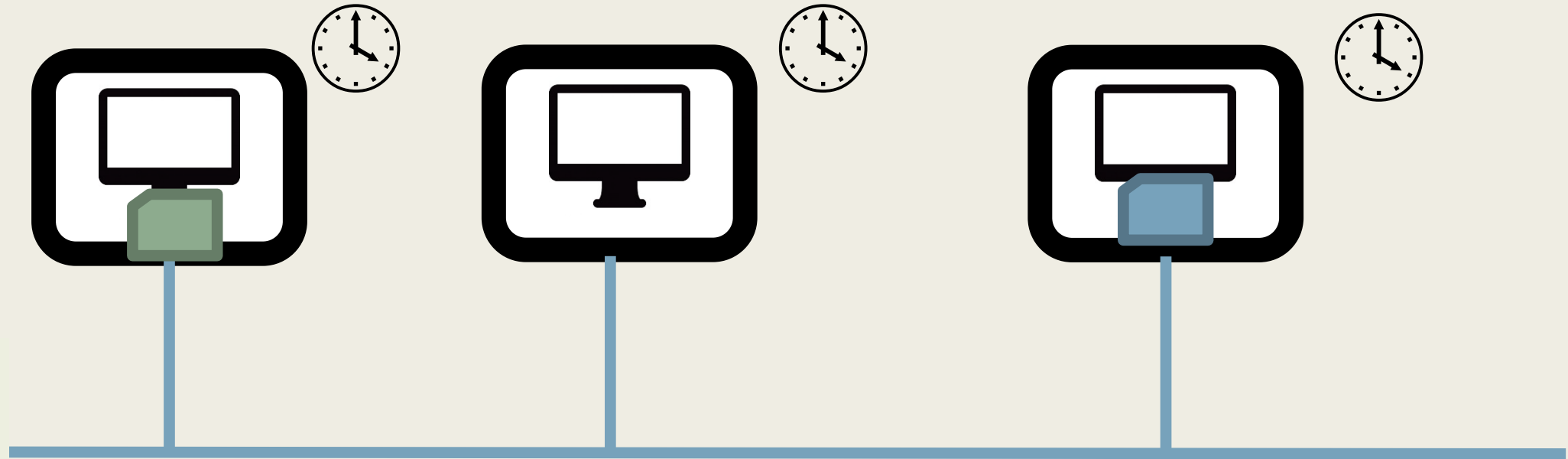
Carrier Sense Multiple Access / Collision Detection



- All hosts hear noise of collision
- Jamming signal sent out

CSMA/CD

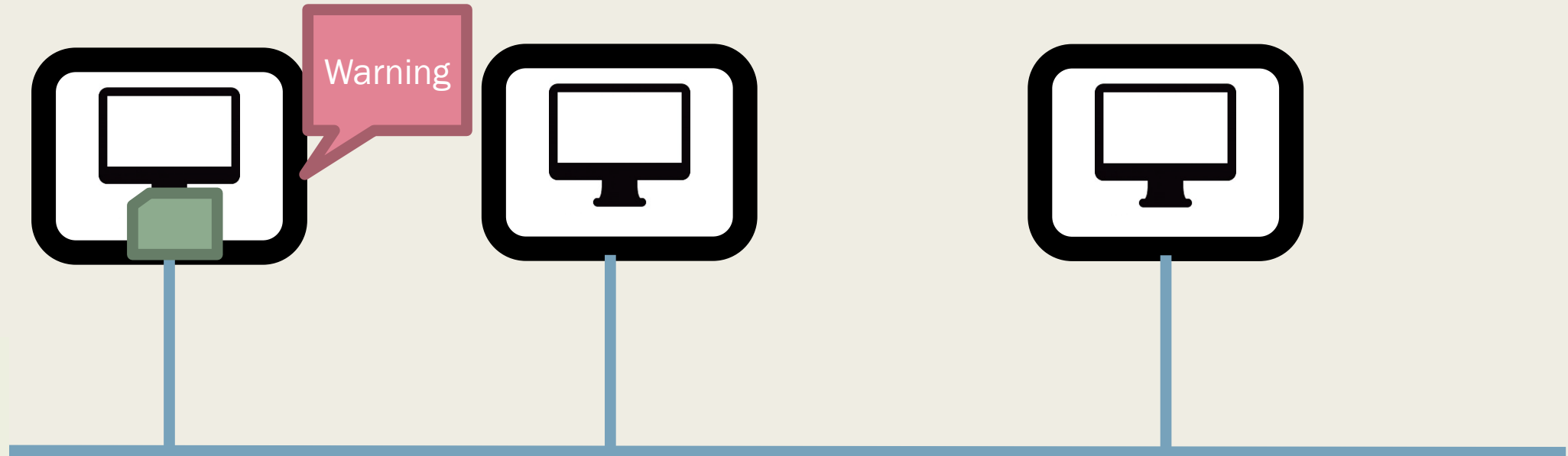
Carrier Sense Multiple Access / Collision Detection



- All hosts wait random length of time before resending packet

CSMA/CA

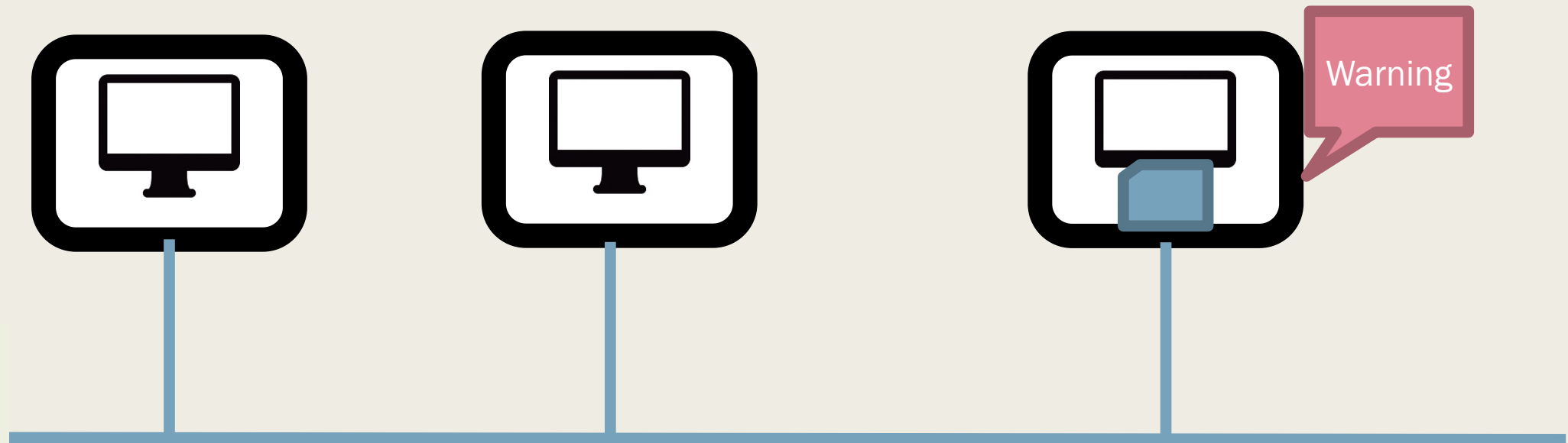
Carrier Sense Multiple Access / Collision Avoidance



- Before sending a message, host sends out a warning.
- When other hosts hear warning, they wait to hear package fly by

CSMA/CA

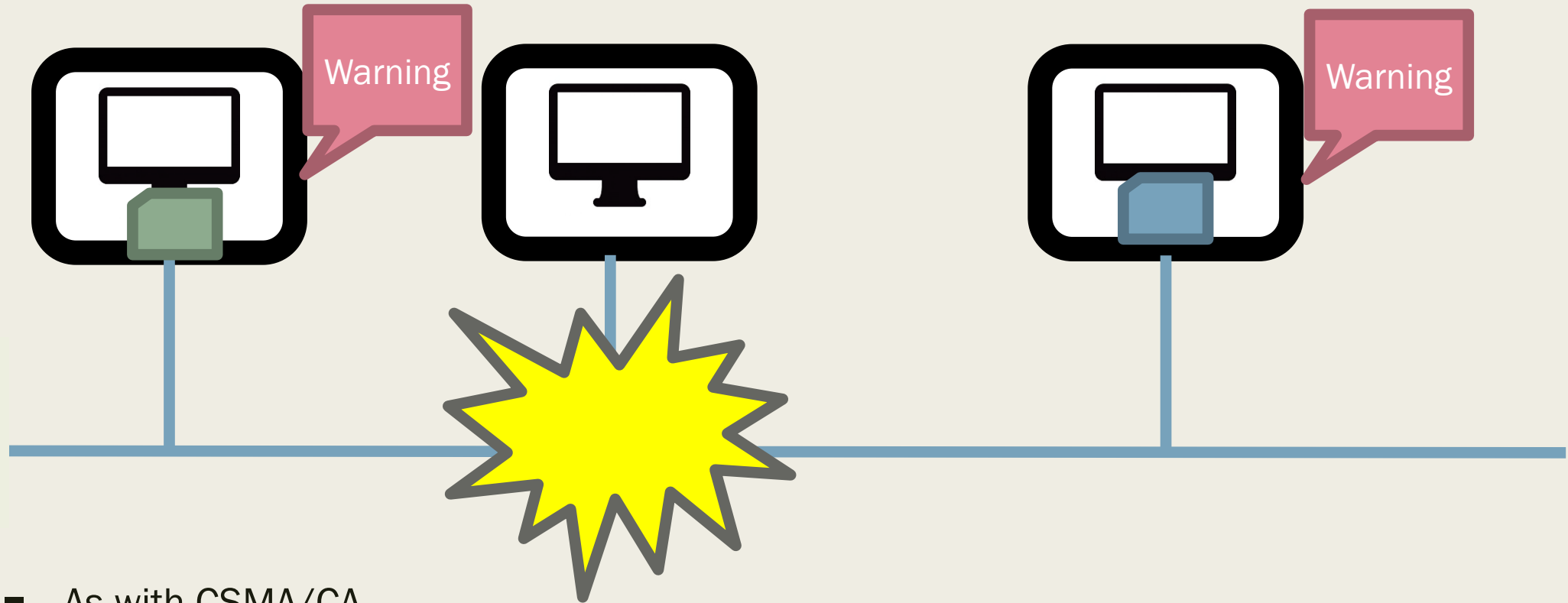
Carrier Sense Multiple Access / Collision Avoidance



- Before sending a message, host sends out a warning.
- When other hosts hear warning, they wait to hear package fly by

CSMA/CA

Carrier Sense Multiple Access / Collision Avoidance



- As with CSMA/CA,
 - *Jamming Signal sent out*
 - *All hosts delay random length*

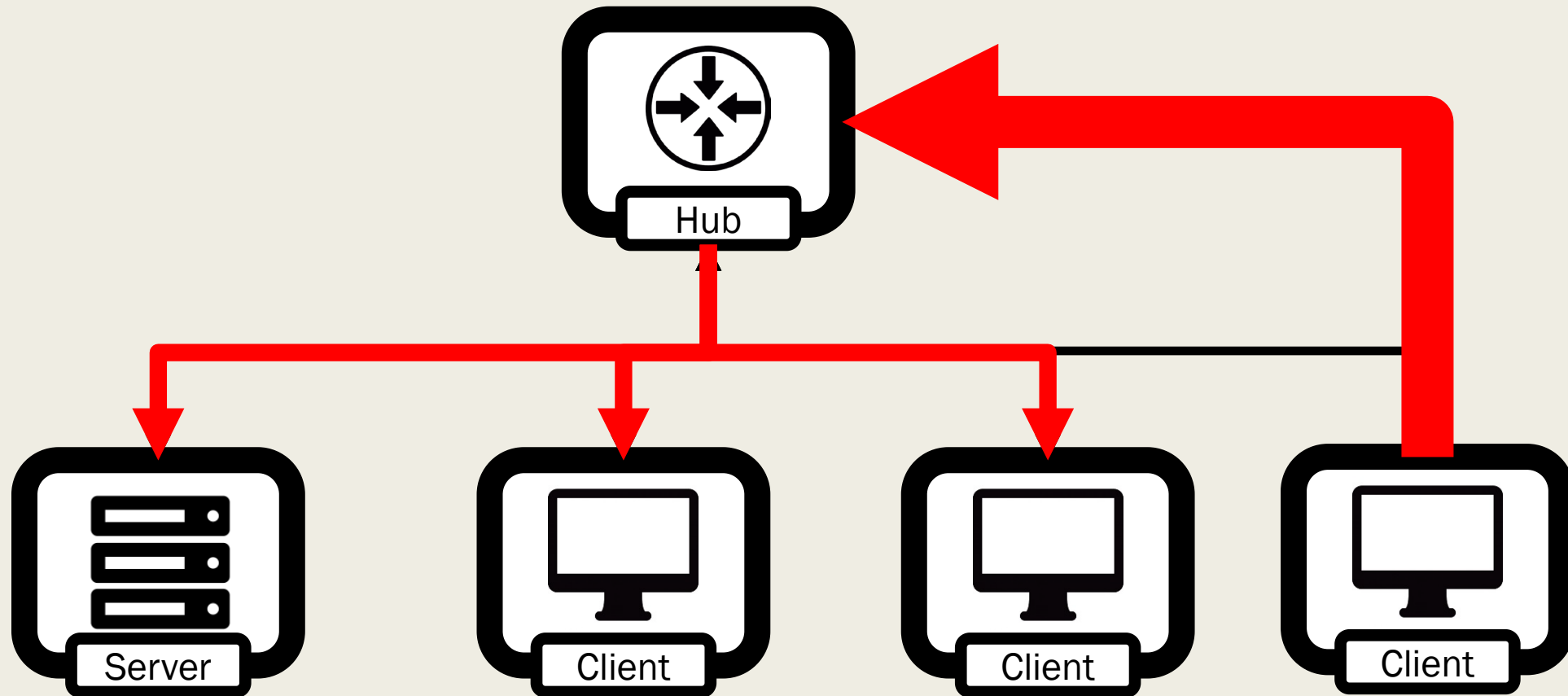
Terminology: Collision Ethernet

- CS
 - *Carrier Sense*
- MA
 - *Multiple Access*
- CD
 - *Collision Detection*
- CA
 - *Collision Avoidance*

NETWORK INFRASTRUCTURE

Hub, Switch, Bridge, Router

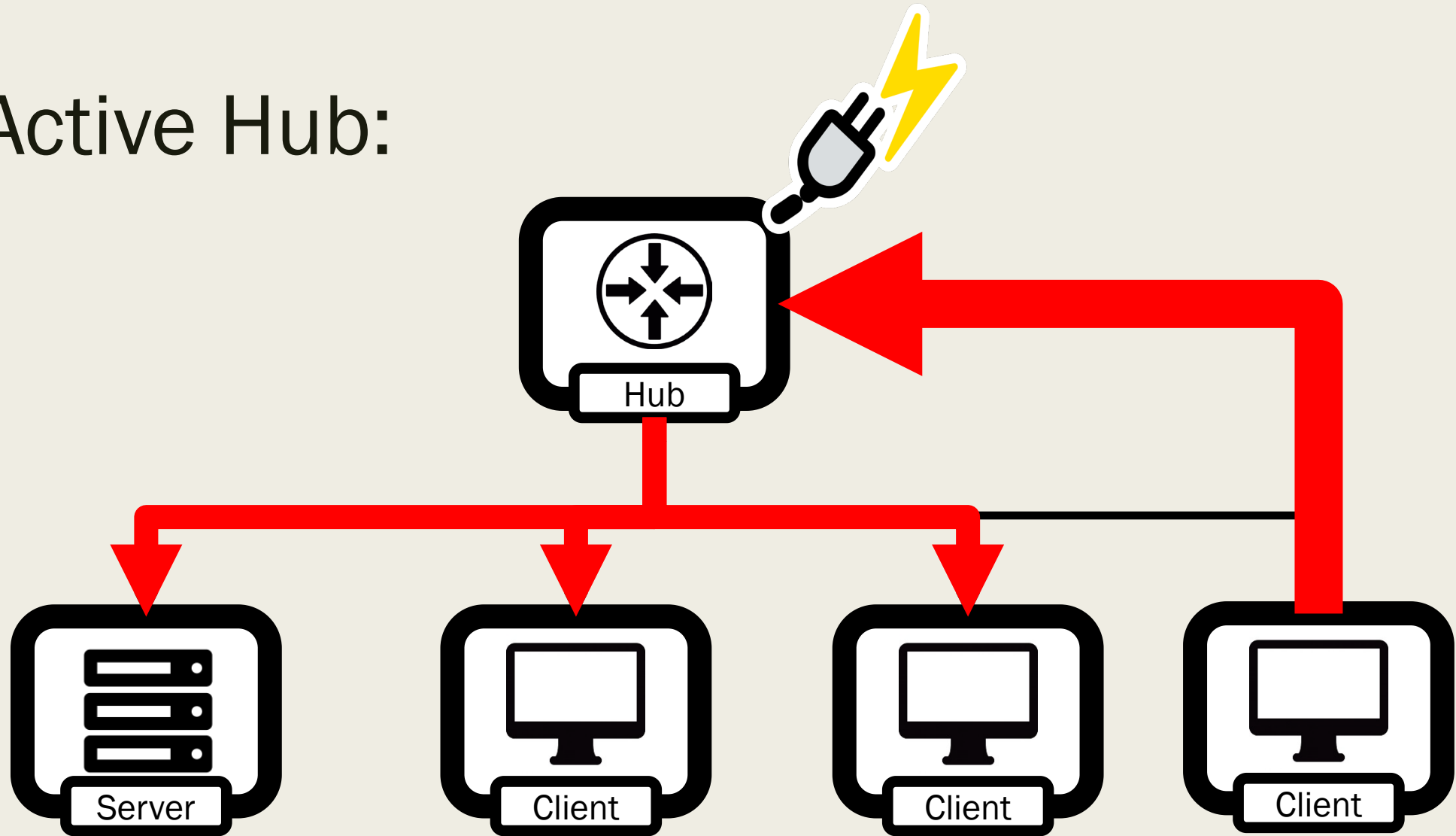
Passive Hub:



Passive Hubs:

- Cable splitter
- Broadcasts any message received to all connected devices
- Bus topology
 - *Risk of collisions*
- No power-source
 - *Signal weakening*

Active Hub:



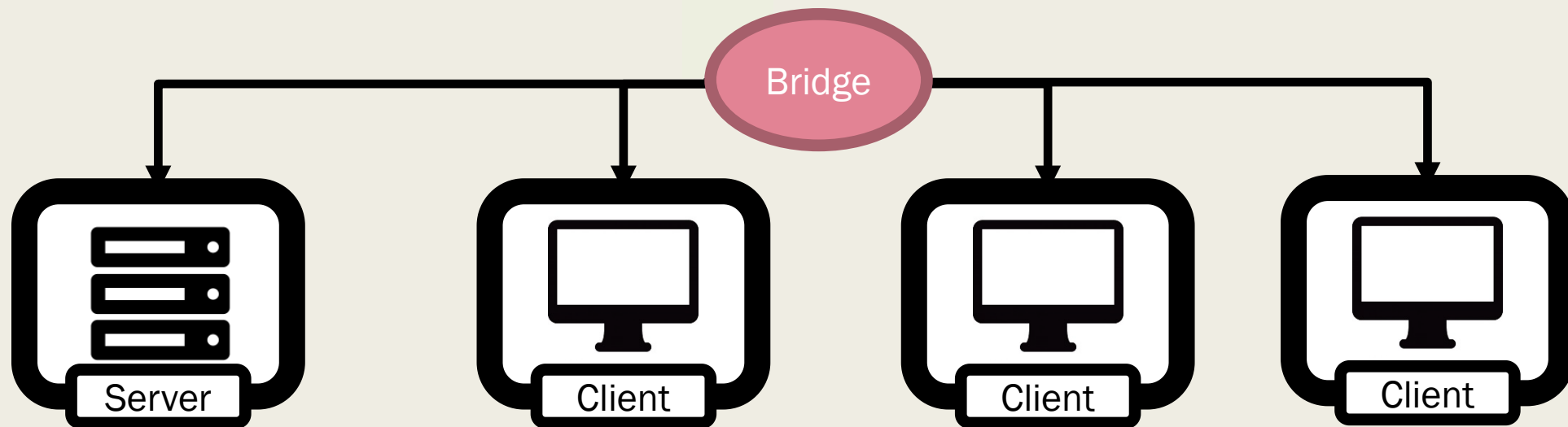
Active Hubs:

- Same broadcast
 - *Bus topology*
 - *Risk of collision*
- But has a power source
 - *Strengthens signal*
- Can be connected to other Active Hubs

Intelligent Hubs:

- Active Hubs with extra functionality
 - *Diagnostics*
 - *Management of ports*

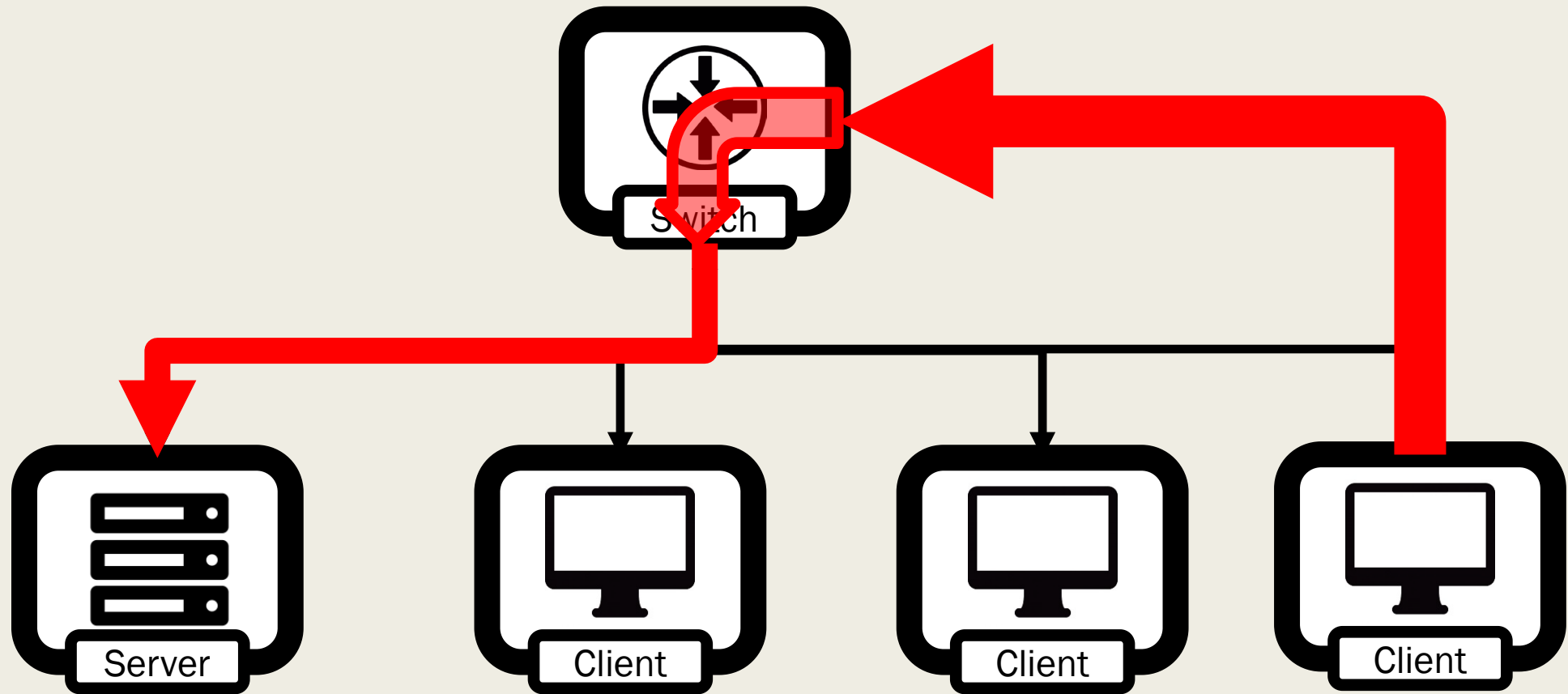
Bridge:



Bridge:

- Breaks up large network into smaller sets of connected hosts:
 - *Collision Domains*
- Will only allow through traffic destined to hosts beyond that Bridge.
- Helps to avoid collisions

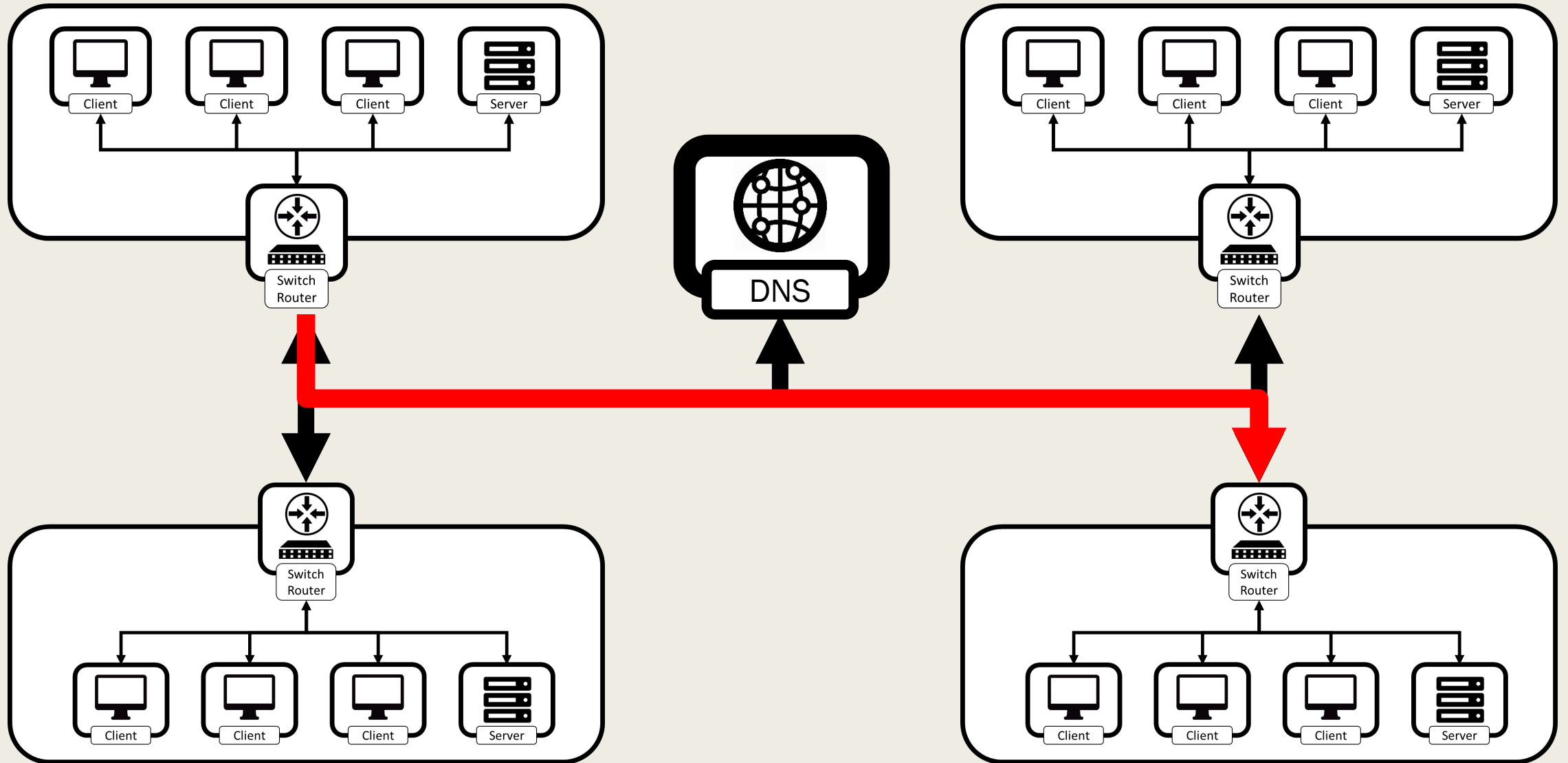
Switch:



Switch:

- Point-to-point connection
- Collision Domain of two machines
- Can also interface between media

Router:



IP ADDRESSING

Networking calculations

IP Addresses:

- Unique Identification of:
 - *Network Host*
 - Source
 - Destination
- Identifies machine's connection to a network
- Moving to another network requires change of IP address
- Assigned by authorities such as:
 - *RIPE (Regional Internet Registry for Europe)*
 - *ARIN (American Registry for Internet Numbers)*
 - *LIR (Local Internet Registries)*
- TCP/IP uses unique 32-bit address
 - *Transmission Control Protocol / Internet Protocol*

IP Addressing, limitations

■ IPv4

- 32 bit address
- Broken into 4 groups of 8 bits
- 2^{32} addresses in total
- 4,294,967,296
- ~2 addresses for every 3 persons on Earth

■ IPv6

- 128 bit address
- Broken into 12 groups of 8 bits
- 2^{128} addresses in total
- $\sim 3.4 \times 10^{38}$
- $\sim 5 \times 10^{28}$ addresses per persons on Earth

Basic structure of an IP v4 address

- 32-bit number (4 octet number ; octet = 8 bits)
 - *Decimal representation:*
 - 133.27.168.125
 - *Binary representation:*
 - 10001010.00011011.10101000.01111101
 - *Hexadecimal representation:*
 - 85.1B.A2.7D

Anatomy of an IP Address:

- Hierarchical Division in IP Address:
 - *Network Part (Prefix)*
 - Describes which physical network
 - *Host Part (Host Address)*
 - Describes which host on that network

205								154								8								1							
1	1	0	0	1	1	0	1	1	0	0	1	1	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1
Network																								Host							

- Boundary can be anywhere
 - *Very often NOT at a multiple of 8 bits*

IP calculations terminology:

- Network Address:
 - *Identifies this network*
- Broadcast Address:
 - *Special IP address used to send a message to all the hosts on this network*
- Valid Host Address:
 - *And IP address that can be allocated to a host in this network*

Three flavours of Network Masks:

- CIDR
 - *Classless Inter-Domain Routing*
 - *Network Prefix*
 - *192.168.1.0/24*
- Network Mask
 - *Bitmask*
 - *255.255.255.0*
- Classful systems

Classless Addressing

- Internet routing and address management today is classless
- CIDR = Classless Inter-Domain Routing
- VLSM = Variable-Length Subnet Masks

Process of Networking calculations:

0 Work out the CIDR

- *The number of bits of the Network Mask*

1. Convert the whole IP address into Binary

2. Network Address is calculated by:

- *Any bits to the left of the Mask, followed by all **zero's** there after*
- *Convert these 4 octet Binary values to Decimal*

3. Broadcast Address is calculated by:

- *Any bits to the left of the Mask, followed by all **one's** there after*
- *Convert these 4 octet Binary values to Decimal*

4. The Network Address and Broadcast Address envelop the range of addressable host IP addresses

- *From the address immediately after the Internet Address*
- *To the address immediately before the Broadcast Address*

Network 10.1.1.32/28

IP Address:

10								•	1								•	1								•	32							
0	0	0	0	1	0	1	0	•	0	0	0	0	0	0	0	1	•	0	0	0	0	0	0	0	1	•	0	0	1	0	0	0	0	0

CIDR /28

Network Address:

0	0	0	0	1	0	1	0	•	0	0	0	0	0	0	0	1	•	0	0	0	0	0	0	0	1	•	0	0	1	0	0	0	0	0
10								•	1								•	1								•	32							

Broadcast Address:

0	0	0	0	1	0	1	0	•	0	0	0	0	0	0	0	1	•	0	0	0	0	0	0	0	1	•	0	0	1	0	1	1	1	1
10								•	1								•	1								•	47							

From:

0	0	0	0	1	0	1	0	•	0	0	0	0	0	0	0	1	•	0	0	0	0	0	0	0	1	•	0	0	1	0	0	0	0	1
10								•	1								•	1								•	33							

Range of
valid Hosts:

To:

0	0	0	0	1	0	1	0	•	0	0	0	0	0	0	0	1	•	0	0	0	0	0	0	0	1	•	0	0	1	0	1	1	1	0
10								•	1								•	1								•	46							

Network address & Broadcast address

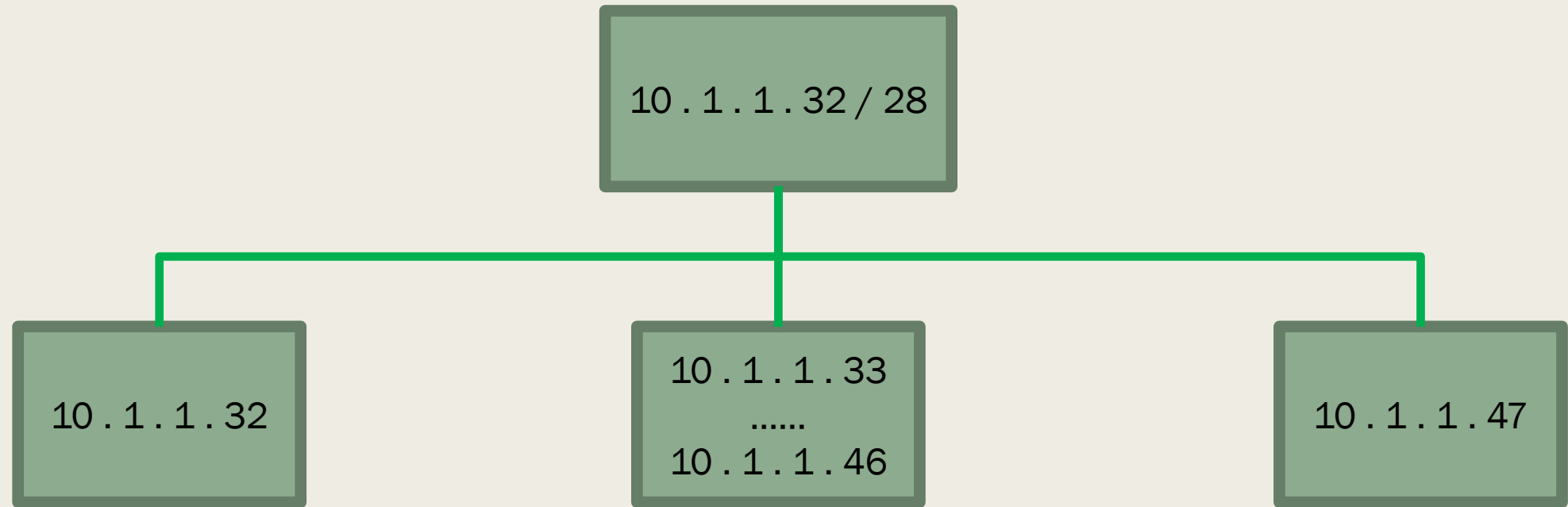
- IP Address with subnet mask defines the range of addresses in the block:

- *10.1.1.32/28 (subnet mask 255.255.255.240)*

IP Address:	00001010	•	00000001	•	00000001	•	00100000
Net Mask:	11111111	•	11111111	•	11111111	•	11110000
Network Ad:	00001010	•	00000001	•	00000001	•	00100000
	10	•	1	•	1	•	32
Broadcast Ad:	00001010	•	00000001	•	00000001	•	00101111
	10	•	1	•	1	•	47

- 10.1.1.32 Network Address (AND operation)
- 10.1.1.47 Broadcast Address
 - *Total of 16 addresses in this subnet;0000 to1111*
- 14 assignable addresses: 10.1.1.33 to 10.1.1.46

Network 10.1.1.32/28



Host: 192.45.63.156/25

IP Address:	<table><tr><td colspan="8">192</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>	192								1	1	0	0	0	0	0	0	•	<table><tr><td colspan="8">45</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr></table>	45								0	0	1	0	1	1	0	1	•	<table><tr><td colspan="8">63</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	63								0	0	1	1	1	1	1	1	•	<table><tr><td colspan="8">156</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>	156								1	0	0	1	1	1	0	0	
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Host: 192.45.63.156 255.255.255.128

IP Address:	192 11000000	•	45 00101101	•	63 00111111	•	156 10011100
Net mask:	255 11111111	•	255 11111111	•	255 11111111	•	128 10000000
Network Address:	11000000 192	•	00101101 45	•	00111111 63	•	10000000 128
Broadcast Address:	11000000 192	•	00101101 45	•	00111111 63	•	11111111 255
From:	11000000 192	•	00101101 45	•	00111111 63	•	10000001 129
To:	11000000 192	•	00101101 45	•	00111111 63	•	11111110 254

Classful networking systems:

- Networks classed by size:
- Class A networks (large):
 - *8 bits network, 24 bits host (/8 , 255.0.0.0)*
 - *First byte in range 0-126*
- Class B Network (medium)
 - *16 bits network, 16 bits host (/16 , 255.255.0.0)*
 - *First byte in range 128-191*
- Class C network (small)
 - *24 bits network, 8 bits host (/24 , 255.255.255.0)*

How to determine what class it is:

- Just look at the IP address:
 - *Class A: 1.0.0.0 to 126.255.255.255*
 - Binary: 0xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
 - 16,777,214 hosts
 - *Class B: 128.0.0.0 to 191.255.255.255*
 - Binary: 10xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
 - 65,534 hosts
 - *Class C: 192.0.0.0 to 223.255.255.255*
 - Binary: 110xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
 - 254 hosts
 - *Class D: (multicast) 224.0.0.0 to 239.255.255.255*
 - Binary: 1110xxxxxxxxxxxxxxxxxxxxxxxxxxxx
 - *Class E: (reserved) 240.0.0.0 to 255.255.255.255*
 - Binary: 1111xxxxxxxxxxxxxxxxxxxxxxxxxxxx

Class table

Class:	Host address range (Den):	Binary (first byte):	CIDR:	Network mask: (den)
A	1 . 0 . 0 . 0 to 126 . 255 . 255 . 255	0 # # # # # # #	/8	255 . 0 . 0 . 0
B	128 . 0 . 0 . 0 to 191 . 255 . 255 . 255	1 0 # # # # # #	/16	255 . 255 . 0 . 0
C	192 . 0 . 0 . 0 to 223 . 255 . 255 . 255	1 1 0 # # # # #	/24	255 . 255 . 255 . 0
D	224 . 0 . 0 . 0 to 239 . 255 . 255 . 255	1 1 1 0 # # # #	(multicast)	
E	240 . 0 . 0 . 0 to 255 . 255 . 255 . 255	1 1 1 1 # # # #	(reserved)	

SUBNETTING

Subnetting calculations:

Maximum number of subnets

Maximum number of hosts per subnet

Traditional Subnetting of Classful Networks:

- Old routing systems allowed a classful network to be divided up into subnets:
 - *All subnets (of one classful network) must be the same size –same netmask*
 - *Subnets cannot be subdivided further*
- None of these restriction apply in modern systems

Class table

Class:	Host address range (Den):	Binary (first byte):	CIDR:	Network mask: (den)
A	1 . 0 . 0 . 0 to 126 . 255 . 255 . 255	0 # # # # # # #	/8	255 . 0 . 0 . 0
B	128 . 0 . 0 . 0 to 191 . 255 . 255 . 255	1 0 # # # # # #	/16	255 . 255 . 0 . 0
C	192 . 0 . 0 . 0 to 223 . 255 . 255 . 255	1 1 0 # # # # #	/24	255 . 255 . 255 . 0
D	224 . 0 . 0 . 0 to 239 . 255 . 255 . 255	1 1 1 0 # # # #	(multicast)	
E	240 . 0 . 0 . 0 to 255 . 255 . 255 . 255	1 1 1 1 # # # #	(reserved)	

The network 193.21.85.0/27

- Maximum number of subnets
- Maximum number of hosts per subnet

- Assume the Network mask is classful
 - *193 is Class C (Network Mask /24)*
- We are told that the Subnet Mask is /27
- Bits available for the Subnets: $27 - 24 = 3$
 - Maximum number of subnets:
 - $2^3 = 8$
- Bits available for the hosts per subnet: $32 - 27 = 5$
 - *Maximum number of hosts per subnet:*
 - $2^5 - 2 = 30$

The network 193.21.85.0/26

- Maximum number of subnets
- Maximum number of hosts per subnet

- Assume the Network mask is classful
 - *193 is Class C (Network Mask /24)*
- We are told that the Subnet Mask is /26
- Bits available for the Subnets: $26 - 24 = 2$
 - Maximum number of subnets:
 - $2^2 = 4$
- Bits available for the hosts per subnet: $32 - 26 = 6$
 - *Maximum number of hosts per subnet:*
 - $2^6 - 2 = 62$

The network 171.21.0.0/22

- Maximum number of subnets
- Maximum number of hosts per subnet

- Assume the Network mask is classful
 - *171 is Class B (Network Mask /16)*
- We are told that the Subnet Mask is /22
- Bits available for the Subnets: $22 - 16 = 6$
 - Maximum number of subnets:
 - $2^6 = 64$
- Bits available for the hosts per subnet: $32 - 22 = 10$
 - *Maximum number of hosts per subnet:*
 - $2^{10} - 2 = 1022$

What we have covered in this unit:

- Network topologies
 - *Physical & Logical*
- Types of network
- Network components
- IP Addressing
 - *Calculations*
 - *Masking*
 - *Classless & Classful systems*
- Subnetting calculations

Thank you

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